

Mohammad Mashrur Khondokar

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EDUCATION

- **Imperial College London** — MSci Mathematics (First Class Honours), 2024

EXPERIENCE

- **Bank of America** — Quantitative Developer, Counterparty Credit Risk (Jul 2024 – Present)
 - Developed and extended exposure models to incorporate **thousands of previously out-of-scope trades**, materially improving risk coverage.
 - Debugged and stabilised a major model release, resolving **99.5% of production issues** through validation and code-level fixes.
 - Developing **repo exposure modelling** and **rates and credit stress-scenario simulations**, focusing on robustness under adverse market conditions.
- **Bank of America** — Software Engineer Intern, Market Risk (Jun 2023 – Aug 2023)
 - Built a production-facing **full-stack analytics application** (Python/Flask, React) on OpenFin for monitoring and querying Market Risk services, including a visualisation dashboard.
 - Designed APIs and data workflows used by risk managers and developers across Market Risk to track critical services.
 - Gained exposure to **VaR models and backtesting pipelines** in a model validation context.

PROJECTS

- **Master's Thesis — Properties of Latent Spaces in Computer Vision Models** (PyTorch)
 - Investigated the geometric structure of latent representations across network depth and training epochs, identifying relationships between latent proximity and semantic similarity.
 - Characterised convexity properties of class-conditioned latent representations and how linear latent combinations could enable future generative control.
- **Sentiment-Driven Trading Strategy** (Python, NLP)
 - Built text-based financial dataset with a pipeline using the Reddit and Yahoo APIs.
 - Fine-tuned a **BERT-based model** for sentiment classification and back-tested an equity trading strategy and identified that performance was largely attributable to the prevailing market regime at the time.

SKILLS

- **Programming:** Python, Julia, JavaScript, GLSL (C-style)
- **ML / Data:** PyTorch, Pandas, NumPy
- **Systems / Web / DevOps:** Flask, React, OpenFin, Material UI, Git, CVS
- **Languages:** English (native), Bengali (native), Japanese (intermediate)